Practice workbook · Sector problems with an unknown radius or angle

Setting up and solving equations in r and theta from given facts about a sector, such as its area and perimeter together, or a sector rolled into a cone.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-TRI-04

1. Question 3 · part whole · 8 marks

2021 NOV · Paper 01

Worked-example retry · Attempt part whole only.

3

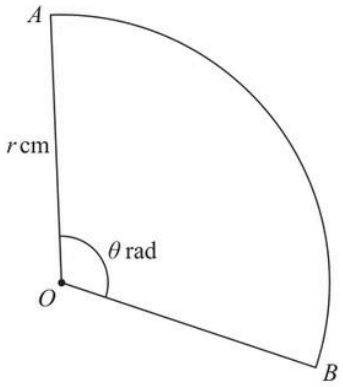


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows the sector AOB of a circle with centre O and radius r cm, where r is an integer. The size of angle AOB is θ radians.

The sector has an area of 16.8 cm^2 and a perimeter of 16.4 cm.

Calculate

(i) the value of r

(ii) the value of θ

(8)

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1. Question 3 · part whole · 8 marks

2021 NOV · Paper 01

Worked-example retry · Attempt part whole only.

Question 3 continued

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(Total for Question 3 is 8 marks)

1. Question 3 · part whole · 8 marks · Working and answer

Working and answer

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2. Question 3 · part whole · 6 marks

2023 JUN · Paper 02

Further practice · Attempt part whole only.

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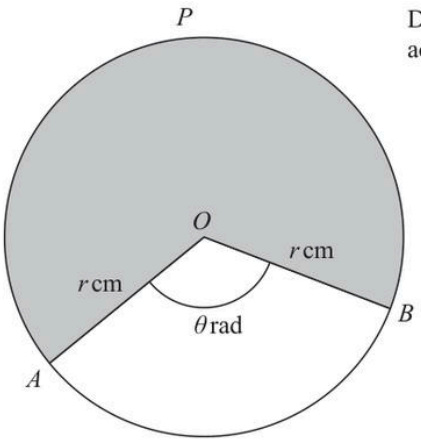


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a circle, centre O, with radius r cm.

The points A, P and B lie on the circle.

The obtuse angle $AOB = \theta$ radians.

The area of the sector $APBO$, shown shaded, is $372.4\,\text{cm}^2$ and the length of the arc APB is 53.2 cm.

Find, to 3 significant figures where appropriate, the value of

(i) r

(ii) θ

(6)

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P 7 2 8 6 6 A 0 6 3 6

2. Question 3 · part whole · 6 marks

2023 JUN · Paper 02

Further practice · Attempt part whole only.

Question 3 continued

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(Total for Question 3 is 6 marks)

2. Question 3 · part whole · 6 marks · Working and answer

Working and answer

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3. Question 4 · part a · 3 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

4

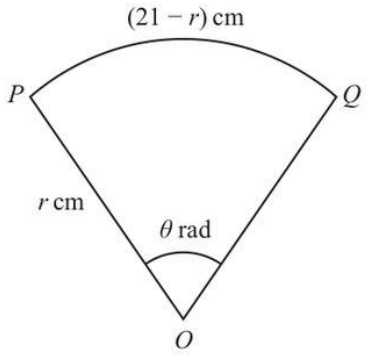


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows the sector OPQ of a circle with centre O and radius r cm.

$OP = OQ = r$ cm arc $PQ = (21 - r)$ cm $\angle POQ = \theta$ radians

The area of the sector is A cm^2

(a) Show that $A = \frac{r}{2}(21 - r)$ **(3)**

The area of the sector must be greater than or equal to 27 cm^2

(b) Find the set of possible values of r **(4)**

(c) Hence write down the set of possible values of θ **(2)**

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P 7 4 0 9 9 A 0 8 3 2

3. Question 4 · part a · 3 marks · Working and answer

Working and answer

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4. Question 4 · part c · 2 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

4

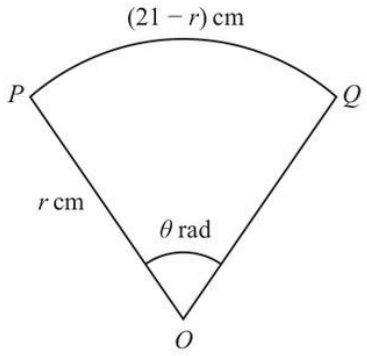


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4. Question 4 · part c · 2 marks

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Further practice · Attempt part c only.

Question 4 continued

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(Total for Question 4 is 9 marks)

4. Question 4 · part c · 2 marks · Working and answer

Working and answer

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5. Question 1 · part whole · 3 marks

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Further practice · Attempt part whole only.

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1 A circle has centre O and radius 14 cm

A sector AOB of the circle has a perimeter of 49 cm

Find, in cm^2, the area of sector AOB

(3)

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(Total for Question 1 is 3 marks)

5. Question 1 · part whole · 3 marks · Working and answer

Working and answer

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6. Question 3 · part a · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part a only.

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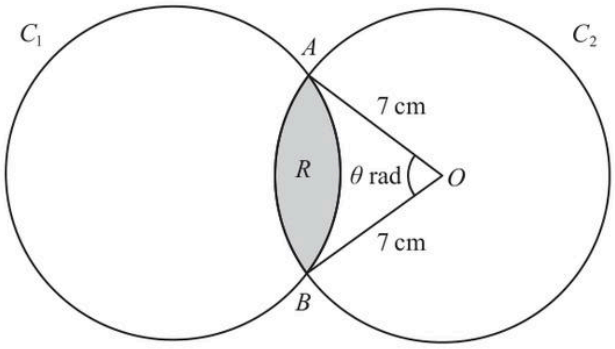


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Figure 1

Figure 1 shows circle C_1 with radius 7 cm and circle C_2 with centre O and radius 7 cm

The circles C_1 and C_2 intersect at the point A and at the point B

The size of angle AOB is θ radians

The perimeter of the region R shown shaded in Figure 1 is $\frac{28\pi}{9}$ cm

(a) Find the exact value of θ **(2)**

(b) Find the area, in cm^2 to 3 significant figures, of the region R **(4)**

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P 7 2 8 6 5 A 0 1 0 3 6

6. Question 3 · part a · 2 marks · Working and answer

Working and answer

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7. Question 10 · part a · 4 marks

2020 NOV · Paper 01

Further practice · Attempt part a only.

10

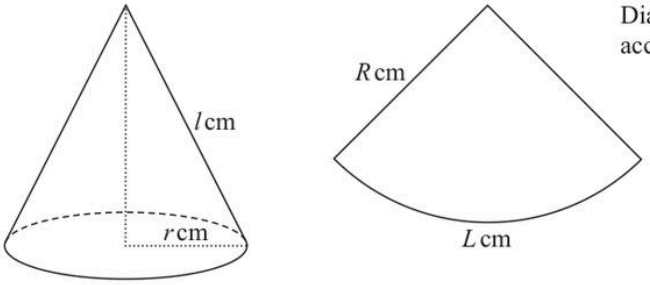


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows a right circular cone with base radius r cm and slant height l cm.
Figure 4 also shows a sector of a circle with radius R cm and arc length L cm.

The area of the curved surface of the cone is A cm^2

By considering how the sector of the circle can be folded to exactly form the curved surface of the cone with R and L suitably chosen,

(a) prove that $A = \pi rl$

(4)

Sand is poured onto a horizontal surface at a constant rate of 1.5 cm^3/s.
The sand forms a pile in the shape of a right circular cone with its base on the surface.
The curved surface area of the cone, A cm^2, increases in such a way that the height of the cone is always three times the radius of the base of the cone.

Given that $\frac{dA}{dr} = k\pi r$, where k is a constant,

(b) find the exact value of k.

(3)

(c) Calculate the rate, in cm^2/s, to 3 significant figures, at which the curved surface area of the pile is increasing when the height of the pile is 24 cm.

(5)

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7. Question 10 · part a · 4 marks · Working and answer

Working and answer

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